Program 1:

Create a table EMPLOYEE using SQL command to store details of employees such as EMPNO, NAME, DESIGNATION, DEPARTMENT, GENDER and SALARY. Specify Primary Key and NOT NULL constraints on the table. Allow only 'M' or 'F' for the column GENDER. DEPARTMENT can be SALES, ACCOUNTS, IT. Choose DESIGNATION as CLERK, ANALYST, MANAGER, ACCOUNTANT and SUPERVISOR that depends on department Write the following SQL queries:

a) Display EMPNO, NAME and DESIGNATION of all employees whose name ends with RAJ.

b) Display the details of all female employees who is earning salary within the range 20000 to 40000 in SALES or IT departments

c) List the different DEPARTMENTs with the DESIGNATIONs in that department.

d) Display the department name, total, average, maximum, minimum salary of the DEPARTMENT only if the total salary given in that department is more than 30000.

e) List the departments which have more than 2 employees.

```
SQL> create table employee(
    empno varchar(10) primary key,
    name varchar(30) not null,
   designation varchar(20) not null check(designation
in('clerk','analyst','manager','accountant','supervisior')),
    department varchar(20) not null check(department in('SALES','ACCOUNTS','IT')),
   gender char(1) not null check(gender in('M','F')),
   salary number(10) not null);

Table created.
```

```
SQL> desc employee;

Name                    Null?          Type
EMPNO                   NOT NULL       VARCHAR2(10)
NAME                    NOT NULL       VARCHAR2(30)
DESIGNATION             NOT NULL       VARCHAR2(20)
DEPARTMENT              NOT NULL       VARCHAR2(20)
GENDER                  NOT NULL       CHAR(1)
SALARY                  NOT NULL       NUMBER(10)


SQL> insert into employee values('A101','Ram','manager','IT','M',12000);
1 row created.
SQL> insert into employee values('A102','Dheeraj','analyst','IT','M',20000);
1 row created.
SQL> insert into employee values('A103','charanraj','clerk','SALES','M',40000);
1 row created.
SQL> insert into employee values('A104','kavya','supervisior','SALES','F',50000);
1 row created.
SQL> insert into employee values('A105','Rakshitha','analyst','IT','F',40000);
1 row created.
SQL> insert into employee values('A106','Ashmitha','clerk','SALES','F',20000);
1 row created.


SQL> select * from employee;

EMPNO   NAME        DESIGNATION    DEPARTMENT   G   SALARY
A101    Ram         manager        IT           M   12000
A102    Dheeraj     analyst        IT           M   20000
A103    charanraj   clerk          SALES        M   40000
A104    kavya       supervisior    SALES        F   50000
A105    Rakshitha   analyst        IT           F   40000
```

A106	Ashmitha	clerk	SALES	F	20000

6 rows selected.

a) SQL> select empno,name,designation from employee where name like'%raj';

EMPNO	NAME	DESIGNATION
A102	Dheeraj	analyst
A103	charanraj	clerk

b) SQL> select * from employee where gender='F' and salary between 20000 and 40000 and department in('SALES','IT');

EMPNO	NAME	DESIGNATION	DEPARTMENT	G	SALARY
A105	Rakshitha	analyst	IT	F	40000
A106	Ashmitha	clerk	SALES	F	20000

c) List the different DEPARTMENTs with the DESIGNATIONs in that department.

SQL> select distinct department ,designation from employee;

DEPARTMENT	DESIGNATION
SALES	clerk
SALES	supervisior
IT	manager
IT	analyst

d)Display the department name, total, average, maximum, minimum salary of the DEPARTMENT only if the total salary given in that department is more than 30000.

SQL> select department,sum(salary)as total_salary,avg(salary)as average_salary,max(salary)as max_salary,min(salary)as min_salary from employee group by department having sum(salary)>30000;

DEPARTMENT	TOTAL_SALARY	AVERAGE_SALARY	MAX_SALARY	MIN_SALARY
IT	72000	24000	40000	12000
SALES	110000	36666.6667	50000	20000

e) List the departments which have more than 2 employees.

SQL> select department from employee group by department having count (*)>2;

DEPARTMENT

IT

SALES

Program 2:

Create a table CLIENT to store CLIENT_NO, NAME, ADDRESS, STATE, BAL_DUE. Client no must start with 'C'. Apply the suitable structure for the columns. Specify Primary Key and NOT NULL constraints on the table. Insert 10 records. Write the following SQL queries:

a) From the table CLIENT, create a new table CLIENT1 that contains only CLIENT_NO and NAME, BAL_DUE from specified STATE. Accept the state during run time.

b) Create a new table CLIENT2 that has the same structure as CLIENT but with no records. Display the structure and records.

c) Add a new column by name PENALTY number (10, 2) to the CLIENT

d) Assign Penalty as 10% of BAL_DUE for the clients C1002, C1005, C1009 and for others 8%. Display Records

e) Change the name of CLIENT1 as NEW_CLIENT

f) Delete the table CLIENT2

```
SQL> create table client(
   client_no varchar(10) primary key check(client_no like 'C%'),
   name varchar(50) not null,
  address varchar(30),
   state varchar(30) not null,
   bal_due number(10,2) not null);


Table created.


SQL> desc client;
Name                          Null?              Type
 CLIENT_NO                    NOT NULL       VARCHAR2(10)
 NAME                         NOT NULL       VARCHAR2(50)
 ADDRESS                                     VARCHAR2(30)
```

STATE	NOT NULL	VARCHAR2(30)
BAL_DUE	NOT NULL	NUMBER(10,2)

```
SQL> insert into client values('C1001','Alice','123 Park lane','Karnataka',1500.00);
1 row created.
SQL> insert into client values('C1002','Bob','456 Hill street','Kerala',2000.00);
1 row created.
SQL> insert into client values('C1003','Charlie','789 Ocean Drive','Tamilnadu',1000.00);
1 row created.
SQL> insert into client values('C1004','Daisy','321 River Road','Karnataka',3000.00);
1 row created.
SQL> insert into client values('C1005','Eva','654 lakeview blvd','Maharastra',2500.00);
1 row created.
SQL> insert into client values('C1006','Frank','888 Sky Ave','Kerala',1200.00);
1 row created.
SQL> insert into client values('C1007','Grace','999 Valley Rd','Karnataka',1800.00);
1 row created.
SQL> insert into client values('C1008','Henry','121 Sunset Blvd','Tamilnadu',2200.00);
1 row created.
SQL> insert into client values('C1009','Ivy','343 green St','Maharastra',1400.00);
1 row created.
SQL> insert into client values('C1010','Jack','676 Forest Lane','Kerala',1600.00);
1 row created.

SQL> select * from client;
```

CLIENT_NO	NAME	ADDRESS	STATE	BAL_DUE
C1001	Alice	123 Park lane	Karnataka	1500
C1002	Bob	456Hill street	Kerala	2000
C1003	Charlie	789 Ocean Drive	Tamilnadu	1000

C1004	Daisy	321 River Road	Karnataka	3000
C1005	Eva	654 lakeview blvd	Maharastra	2500
C1006	Frank	888 Sky Ave	Kerala	1200
C1007	Grace	999 Valley Rd	Karnataka	1800
C1008	Henry	121 Sunset Blvd	Tamilnadu	2200
C1009	Ivy	343 green St	Maharastra	1400
C1010	Jack	676 Forest Lane	Kerala	1600

10 rows selected.

a) From the table CLIENT, create a new table CLIENT1 that contains only CLIENT_NO and NAME, BAL_DUE from specified STATE. Accept the state during run time.

SQL> Accept state_input char prompt'enter state:'

enter state:karnataka

SQL> create table client1 as select client_no,name,bal_due from client where state='&state_input';

old 1: create table client1 as select client_no,name,bal_due from client where state='&state_input'

new 1: create table client1 as select client_no,name,bal_due from client where state='karnataka'

Table created.

b) Create a new table CLIENT2 that has the same structure as CLIENT but with no records. Display the structure and records.

SQL> create table client2 as select * from client where 1=0;

Table created.

SQL> desc client2;

```
Name                                      Null?    Type
----------------------------------------- -------- ----------------------------
CLIENT_NO                                          VARCHAR2(10)
NAME                                      NOT NULL VARCHAR2(50)
```

ADDRESS		VARCHAR2(30)
STATE	NOT NULL	VARCHAR2(30)
BAL_DUE	NOT NULL	NUMBER(10,2)

SQL> select * from client2;

no rows selected

c) Add a new column by name PENALTY number (10, 2) to the CLIENT

SQL> alter table client add penalty number (10,2);

Table altered.

SQL> desc client;

Name	Null?	Type
CLIENT_NO	NOT NULL	VARCHAR2(10)
NAME	NOT NULL	VARCHAR2(50)
ADDRESS		VARCHAR2(30)
STATE	NOT NULL	VARCHAR2(30)
BAL_DUE	NOT NULL	NUMBER(10,2)
PENALTY		NUMBER(10,2)

d) Assign Penalty as 10% of BAL_DUE for the clients C1002, C1005, C1009 and for others 8%. Display Records

SQL> update client set penalty=case when client_no in('C1002','C1005','C1009')then bal_due*0.10

else bal_due*0.08

end;

10 rows updated.

SQL> select * from client;

CLIENT_NO	NAME	ADDRESS	STATE	BAL_DUE	PENALTY
C1001	Alice	123 Park lane	Karnataka	1500	120
C1002	Bob	456 Hill street	Kerala	2000	200
C1003	Charlie	789 Ocean Drive	Tamilnadu	1000	80
C1004	Daisy	321 River Road	Karnataka	3000	240
C1005	Eva	654 lakeview	Maharastra	2500	250
C1006	Frank	888 Sky Ave	Kerala	1200	96
C1007	Grace	999 Valley Rd	Karnataka	1800	144
C1008	Henry	121 Sunset Blvd	Tamilnadu	2200	176
C1009	Ivy	343 green St	Maharastra	1400	140
C1010	Jack	676 Forest Lane	Kerala	1600	128

10 rows selected.

e) Change the name of CLIENT1 as NEW_CLIENT

SQL> alter table client1 rename to new_client;

Table altered.

f) Delete the table CLIENT2

SQL> drop table client2;

Table dropped.

Program 3:

Create a table BOOK using SQL command to store Accession No, TITLE, AUTHOR, PUBLISHER, YEAR, PRICE. Apply the suitable structure for the columns. Specify Primary Key and NOT NULL constraints on the table. Insert 10 records. Write the following SQL queries:

a) List the details of publishers having 'a' as the second character in their names.

b) Display Accession No., TITLE, PUBLISHER and YEAR of the books published by the specified author before 2010 in the descending order of YEAR. Accept author during run time

c) Modify the size of TITLE to increase the size 5 characters more.

d) Display the details of all books other than Microsoft press publishers.

e) Remove the records of the books published before 1990.

```
SQL> create table Book(
  2  AccessionNo varchar2(10) primary key,
  3  Title varchar2(50) not null,
  4  Author varchar2(50) not null,
  5  Publisher varchar2(50) not null,
  6  Year number(4) not null,
  7  Price number(8,2) not null);

Table created.

SQL> desc Book;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ACCESSIONNO                               NOT NULL VARCHAR2(10)
 TITLE                                     NOT NULL VARCHAR2(50)
 AUTHOR                                    NOT NULL VARCHAR2(50)
```

```
PUBLISHER                    NOT NULL        VARCHAR2(50)
YEAR                         NOT NULL        NUMBER(4)
PRICE                        NOT NULL        NUMBER(8,2)


SQL> insert into Book values('B001','Learning Sql','Alan Beaulieu','Oreilly',2009,450.00);

1 row created.

SQL> insert into Book values('B002','Sql in 10 minutes','Ben Forta','Sams',2012,550.00);

1 row created.

SQL> insert into Book values('B003','Head First Sql','lynn beighley','Oreilly',2007,600.00);

1 row created.

SQL> insert into Book values('B004','data base system','Thomas connolly','Pearson',2010,750.00);

1 row created.

SQL> insert into Book values('B005','Sql for dummies','Allen taylor','wiley',2008,400.00);

1 row created.

SQL> insert into Book values('B006','Pro Sql server','Adam machanic','Apress',2013,850.00);

1 row created.

SQL> insert into Book values('B007','T-sqlSql Fundamentals','Itzik Ben-Gan','Microsoft press',2009,670.00);

1 row created.

SQL> insert into Book values('B008','MySql CookBook','Paul Dubois','Oreilly',2011,580.00);

1 row created.

SQL> insert into Book values('B009','EffectiveSql','John Viescas','Addison-Wesley',2017,500.00);

1 row created.

SQL> insert into Book values('B010','Beginning Sql','Paul Wilton','Wrox',2003,430.00);

1 row created.
```

SQL> select * from Book;

ACCESSIONN	TITLE	AUTHOR	PUBLISHER	YEAR	PRICE
B001	Learning Sql	Alan Beaulieu	Oreilly	2009	450
B002	Sql in 10 minutes	Ben Forta	Sams	2012	550
B003	Head First Sql	lynn beighley	Oreilly	2007	600
B004	data base system	Thomas Connolly	Pearson	2010	750
B005	Sql for dummies	Allen taylor	wiley	2008	400
B006	Pro Sql server	Adam machanic	Apress	2013	850
B007	T-sqlSql Fundamentals	Itzik Ben-Gan	Microsoft press	2009	670
B008	MySql CookBook	Paul Dubois	Oreilly	2011	580
B009	EffectiveSql	John Viescas	Addison-Wesley	2017	500
B010	Beginning Sql	Paul Wilton	Wrox	2003	430

10 rows selected.

a) List the details of publishers having 'a' as the second character in their names.

SQL> select * from Book where publisher like'_a%';

ACCESSIONN	TITLE	AUTHOR	PUBLISHER	YEAR	PRICE
B002	Sql in 10 minutes	BenForta	Sams	2012	550

b) Display Accession No., TITLE, PUBLISHER and YEAR of the books published by the specified author before 2010 in the descending order of YEAR. Accept author during run time

SQL> select AccessionNo,Title,Publisher,Year from Book where Author='Alan Beaulieu' and year<2010 order by Year desc;

ACCESSIONN	TITLE	PUBLISHER	YEAR
B001	Learning Sql	Oreilly	2009

c) Modify the size of TITLE to increase the size 5 characters more.

SQL> alter table Book modify Title varchar2(55);

Table altered.

SQL> desc Book;

Name	Null?	Type
ACCESSIONNO	NOT NULL	VARCHAR2(10)
TITLE	NOT NULL	VARCHAR2(55)
AUTHOR	NOT NULL	VARCHAR2(50)
PUBLISHER	NOT NULL	VARCHAR2(50)
YEAR	NOT NULL	NUMBER(4)
PRICE	NOT NULL	NUMBER(8,2)

d) Display the details of all books other than Microsoft press publishers.

SQL> select * from book where publisher<>'Microsoft press';

ACCESSIONN	TITLE	AUTHOR	PUBLISHER	YEAR	PRICE
B001	Learning Sql	Alan Beaulieu	Oreilly	2009	450
B002	Sql in 10 minutes	Ben Forta	Sams	2012	550
B003	Head First Sql	lynn beighley	Oreilly	2007	600
B004	data base system	Thomas Connolly	Pearson	2010	750
B005	Sql for dummies	Allen taylor	wiley	2008	400
B006	Pro Sql server	Adam machanic	Apress	2013	850
B008	MySql CookBook	Paul Dubois	Oreilly	2011	580
B009	EffectiveSql	John Viescas	Addison-Wesley	2017	500
B010	Beginning Sql	Paul Wilton	Wrox	2003	430

09 rows selected.

e) Remove the records of the books published before 1990

SQL> delete from book where year<1990;

0 rows deleted.

Program 4:

Create a table SALES with columns SNO, SNAME, MNO, JOIN_DATE, DATE_BIRTH, SALARY, SALES_AMOUNT and COMMISSION. Minimum age for joining the company must be 18 Yrs. Default value for Commission should be 0. Apply the suitable structure for the columns. Specify Primary Key and NOT NULL constraints on the table. Insert 10 records with data except commission. Manager of Manager can be NULL. Write the following SQL queries:

a) Display the details of Sales Persons whose salary is more than Average salary in the company.

b) Update commission as 20% of Sales Amount.

c) Display SNO, SNAME, MNO, SALARY, COMMISSION, MANAGER_SALARY of the sales persons getting sum of salary and commission more than salary of manager. (Self-join)

d) Display the records of employees who finished the service of 10years

```
SQL> create table sales(
   sno number(4) primary key,
   sname varchar2(30) not null,
   manager_name varchar2(30),
   join_date date not null,
   date_birth date not null,
   salary number(10,2) not null,
   sales_amount number(10,2) not null,
   commission number(10,2) default 0 check(commission>=0),
  CONSTRAINT age_check CHECK(months_between(join_date,date_birth)>=216));


Table created.
```

```
SQL> desc sales;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                       NOT NULL NUMBER(4)
 SNAME                                     NOT NULL VARCHAR2(30)
 MANAGER_NAME                                       VARCHAR2(30)
 JOIN_DATE                                 NOT NULL DATE
 DATE_BIRTH                                NOT NULL DATE
 SALARY                                    NOT NULL NUMBER(10,2)
 SALES_AMOUNT                              NOT NULL NUMBER(10,2)
 COMMISSION                                         NUMBER(10,2)

SQL> insert intosales(sno,sname,manager_name,join_date,date_birth,salary,sales_amount)values
  (1,'john','peter',TO_DATE('2015-06-15','YYYY-MM-DD'),TO_DATE('1990-05-20','YYYY-MM-DD'),50000,300000);

1 row created.

SQL> insert into sales(sno,sname,manager_name,join_date,date_birth,salary,sales_amount)values
   (2,'alice','peter',TO_DATE('2017-01-10','YYYY-MM-DD'),TO_DATE('1992-04-12','YYYY-MM-DD'),48000,250000);

1 row created.

SQL> insert into sales(sno,sname,manager_name,join_date,date_birth,salary,sales_amount)values
   (3,'david','john',TO_DATE('2016-09-25','YYYY-MM-DD'),TO_DATE('1991-11-01','YYYY-MM-DD'),47000,220000);

1 row created.
```

```
SQL> insert into
sales(sno,sname,manager_name,join_date,date_birth,salary,sales_amount)values

   (4,'sophia','alice',TO_DATE('2018-07-05','YYYY-MM-DD'),TO_DATE('1994-02-
18','YYYY-MM-DD'),46000,210000);

1 row created.


SQL> insert into
sales(sno,sname,manager_name,join_date,date_birth,salary,sales_amount)values

   (5,'liam','john',TO_DATE('2020-03-01','YYYY-MM-DD'),TO_DATE('1996-08-
25','YYYY-MM-DD'),45000,190000);

1 row created.


SQL> insert into
sales(sno,sname,manager_name,join_date,date_birth,salary,sales_amount)values

  (6,'Emma','peter',TO_DATE('2014-12-10','YYYY-MM-DD'),TO_DATE('1988-07-
14','YYYY-MM-DD'),55000,330000);

1 row created.


SQL> insert into
sales(sno,sname,manager_name,join_date,date_birth,salary,sales_amount)values

   (7,'noah','david',TO_DATE('2019-04-20','YYYY-MM-DD'),TO_DATE('1995-03-
02','YYYY-MM-DD'),44000,180000);

1 row created.


SQL> insert into
sales(sno,sname,manager_name,join_date,date_birth,salary,sales_amount)values

  (8,'olivia','sophia',TO_DATE('2021-06-30','YYYY-MM-DD'),TO_DATE('2000-01-
01','YYYY-MM-DD'),43000,175000);

1 row created


SQL> insert into
sales(sno,sname,manager_name,join_date,date_birth,salary,sales_amount)values
```

(9,'william','liam',TO_DATE('2013-11-15','YYYY-MM-DD'),TO_DATE('1985-10-10','YYYY-MM-DD'),52000,295000);

1 row created.

SQL> insert into sales(sno,sname,manager_name,join_date,date_birth,salary,sales_amount)values

(10,'ava','null',TO_DATE('2012-08-01','YYYY-MM-DD'),TO_DATE('1980-05-05','YYYY-MM-DD'),60000,350000);

1 row created.

SQL> select * from sales;

SNO	SNAME	MANAGER_NAME	JOIN_DATE	DATE_BIRT	SALARY	SALES_AMOUNT	COMMISSION
1	john	peter	15-JUN-15	20-MAY-90	50000	300000	0
2	alice	peter	10-JAN-17	12-APR-92	48000	250000	0
3	david	john	25-SEP-16	01-NOV-91	47000	220000	0
4	sophia	alice	05-JUL-18	18-FEB-94	46000	210000	0
5	liam	john	01-MAR-20	25-AUG-96	45000	190000	0
6	Emma	peter	10-DEC-14	14-JUL-88	55000	330000	0
7	noah	David	20-APR-19	02-MAR-95	44000	180000	0
8	olivia	Sophia	30-JUN-21	01-JAN-00	43000	175000	0
9	william	liam	15-NOV-13	10-OCT-85	52000	295000	0
10	ava	null	01-AUG-12	05-MAY-80	60000	350000	0

10 rows selected.

a) Display the details of Sales Persons whose salary is more than Average salary in the company.

SQL> select * from sales where salary>(select avg(salary)from sales);

SNO	SNAME	MANAGER_NAME	JOIN_DATE	DATE_BIRT	SALARY	SALES_AMOUNT	COMMISSION
1	john	peter	15-JUN-15	20-MAY-90	50000	300000	0
6	Emma	peter	10-DEC-14	14-JUL-88	55000	330000	0
9	william	liam	15-NOV-13	10-OCT-85	52000	295000	0
10	ava	null	01-AUG-12	05-MAY-80	60000	350000	0

b) Update commission as 20% of Sales Amount.

SQL> update sales set commission =sales_amount*0.20;

10 rows updated.

SQL> select * from sales;

SNO	SNAME	MANAGER_NAME	JOIN_DATE	DATE_BIRT	SALARY	SALES_AMOUNT	COMMISSION
1	john	peter	15-JUN-15	20-MAY-90	50000	300000	60000
2	alice	peter	10-JAN-17	12-APR-92	48000	250000	50000
3	david	john	25-SEP-16	01-NOV-91	47000	220000	44000
4	sophia	alice	05-JUL-18	18-FEB-94	46000	210000	42000
5	liam	john	01-MAR-20	25-AUG-96	45000	190000	38000
6	Emma	peter	10-DEC-14	14-JUL-88	55000	330000	66000
7	noah	David	20-APR-19	02-MAR-95	44000	180000	36000
8	olivia	Sophia	30-JUN-21	01-JAN-00	43000	175000	35000
9	william	liam	15-NOV-13	10-OCT-85	52000	295000	59000
10	ava	null	01-AUG-12	05-MAY-80	60000	350000	70000

10 rows selected.

d) Display the records of employees who finished the service of 10years

SQL> select * from sales where add_months(join_date,120)<=sysdate;

SNO	SNAME	MANAGER_NAME	JOIN_DATE	DATE_BIRT	SALARY	SALES_AMOUNT	COMMISSION
1	john	peter	15-JUN-15	20-MAY-90	50000	300000	60000
6	Emma	peter	10-DEC-14	14-JUL-88	55000	330000	66000
9	william	liam	15-NOV-13	10-OCT-85	52000	295000	59000
10	ava	null	01-AUG-12	05-MAY-80	60000	350000	70000

c) Display SNO, SNAME, MNO, SALARY, COMMISSION, MANAGER_SALARY of the sales persons getting sum of salary and commission more than salary of manager. (Self-join)

SQL> select e.sno,e.sname,e.manager_name,e.salary,e.commission,m.salary as manager_salary from sales e join sales m on e.manager_name=m.sname where(e.salary+e.commission)>m.salary;

SNO	SNAME	MANAGER_NAME	SALARY	COMMISSION	MANAGER_SALARY
5	liam	john	45000	38000	50000
3	david	john	47000	44000	50000
4	sophia	alice	46000	42000	48000
7	noah	david	44000	36000	47000
8	olivia	Sophia	43000	35000	46000
9	william	liam	52000	59000	45000

6 rows selected.

Program 5:

Create a table Sales_Details with the columns SNO, MONTH, TARGET and QTY_SOLD to store the Sales Details of one year. Specify the composite primary key to the columns SNO and MONTH. TARGET and SALES must be positive numbers. Write the following SQL queries:

a) Display the total sales by each sales person considering only those months sales where target was reached

b) If a commission of RS.50 provided for each item after reaching target, calculate and display the total commission for each sales person.

c) Display the SNO of those who never reached the target.

d) Display the SNO, MONTH and QTY_SOLD of the sales persons with SNO S0001 or S0003

```
SQL> create table Sales_Details(
  2  sno varchar(10),
  3  month varchar(10),
  4  target number(10,2)check(target>0),
  5  qty_sold number(10)check(qty_sold>0),
  6  primary key(sno,month));
Table created.


SQL> desc Sales_Details;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SNO                                       NOT NULL VARCHAR2(10)
 MONTH                                     NOT NULL VARCHAR2(10)
 TARGET                                             NUMBER(10,2)
 QTY_SOLD                                           NUMBER(10)
```

```
SQL> insert into Sales_Details values('S0001','Jan',100,120);

1 row created.

SQL> insert into Sales_Details values('S0001','Feb',150,130);

1 row created.

SQL> insert into Sales_Details values('S0002','Jan',100,90);

1 row created.

SQL> insert into Sales_Details values('S0002','Feb',150,160);

1 row created.

SQL> insert into Sales_Details values('S0003','Jan',120,130);

1 row created.

SQL> insert into Sales_Details values('S0003','Feb',100,80);

1 row created.


SQL> select * from Sales_Details;

SNO      MONTH      TARGET   QTY_SOLD

S0001      Jan         100      120

S0001      Feb         150      130

S0002      Jan         100       90

S0002      Feb         150      160

S0003      Jan         120      130

S0003      Feb         100       80

6 rows selected.
```

a) Display the total sales by each sales person considering only those months sales where target was reached

```
SQL> select sno,sum(qty_sold)as total_sales from Sales_Details where qty_sold>=target
group by sno;

SNO      TOTAL_SALES

---------- -----------

S0001        120
```

```
S0002          160

S0003          130
```

b) If a commission of RS.50 provided for each item after reaching target, calculate and display the total commission for each sales person.

```
SQL> select sno,sum((qty_sold-target) *50) as total_commission from Sales_Details where qty_sold>target group by sno;

SNO       TOTAL_COMMISSION

---------- ----------------

S0001              1000

S0002               500

S0003               500
```

c) Display the SNO of those who never reached the target.

```
SQL> select sno from Sales_Details group by sno having max(qty_sold) <min(target);

no rows selected
```

d) Display the SNO, MONTH and QTY_SOLD of the sales persons with SNO S0001 or S0003.

```
SQL> select sno,month,qty_sold from Sales_Details where sno in('S0001','S0003');

SNO       MONTH       QTY_SOLD

---------- ---------- ----------

S0001     Feb            130

S0001     Jan            120

S0003     Feb             80

S0003     Jan            130
```

Program 6:

Create the following tables by identifying primary and foreign keys. Specify the not null property for mandatory keys. SUPPLIERS (SUPPLIER_NO, SNAME, SADDRESS, SCITY) COMPUTER_ITEMS (ITEM_NO, SUPPLIER_NO, ITEM_NAME, IQUANTITY) Consider three suppliers. A supplier can supply more than one type of items. Write the SQL queries for the following:

a) List ITEM and SUPPLIER details in alphabetical order of city name and in each city decreasing order of IQUANTITY.

b) List the name, city,and address of the suppliers who are supplying keyboard.

c) List the supplier's name, items supplied by the suppliers 'Cats' and 'Electrotech'.

d) Find the items having quantity less than 5 and insert the details of supplier and item of these, into another table NEWORDER

```
SQL> create table suplier(
  2  supplier_no varchar(10) primary key,
  3  sname varchar(50) not null,
  4  saddress varchar(100),
  5  scity varchar(50) not null);
Table created.


SQL> desc suplier;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SUPPLIER_NO                               NOT NULL VARCHAR2(10)
 SNAME                                     NOT NULL VARCHAR2(50)
 SADDRESS                                           VARCHAR2(100)
 SCITY                                     NOT NULL VARCHAR2(50)
```

```
SQL> create table computer_items(
  2  item_no varchar(10) primary key,
  3  supplier_no varchar(10) not null,
  4  item_name varchar(50) not null,
  5  Iquantity number(5)check(Iquantity>=0),
  6  Foreign key(supplier_no)references suplier(supplier_no));
Table created.

SQL> desc computer_items;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ITEM_NO                                   NOT NULL VARCHAR2(10)
 SUPPLIER_NO                               NOT NULL VARCHAR2(10)
 ITEM_NAME                                 NOT NULL VARCHAR2(50)
 IQUANTITY                                          NUMBER(5)

SQL> insert into suplier values('S1','cats','12 main st','Delhi');
1 row created.
SQL> insert into suplier values('S2','electrotech','45 tech ave','Mumbai');
1 row created.
SQL> insert into suplier values('S3','digitech','78 supply rd','Delhi');
1 row created.
SQL> select * from suplier;
SUPPLIER_N   SNAME        SADDRESS       SCITY
S1           cats         12 main st     Delhi
S2           electrotech  45 tech ave    Mumbai
S3           digitech     78 supply rd   Delhi
```

SQL> insert into computer_items values('I1','S1','Mouse',10);

1 row created.

SQL> insert into computer_items values('I2','S1','Keyboard',3);

1 row created.

SQL> insert into computer_items values('I3','S2','Monitor',15);

1 row created.

SQL> insert into computer_items values('I4','S2','Keyboard',2);

1 row created.

SQL> insert into computer_items values('I5','S3','CPU',8);

1 row created.

SQL> select * from computer_items;

ITEM_NO	SUPPLIER_N	ITEM_NAME	IQUANTITY
I1	S1	Mouse	10
I2	S1	Keyboard	3
I3	S2	Monitor	15
I4	S2	Keyboard	2
I5	S3	CPU	8

a) List ITEM and SUPPLIER details in alphabetical order of city name and in each city decreasing order of IQUANTITY.

SQL> select ci.item_name,ci.Iquantity,s.supplier_no,s.sname,s.saddress,s.scity from computer_items ci join suplier s on ci.supplier_no=s.supplier_no order by s.scity asc,ci.Iquantity desc;

ITEM_NAME	IQUANTITY	SUPPLIER_N	SNAME	SADDRESS	SCITY
Mouse	10	S1	cats	12 main st	Delhi
CPU	8	S3	digitech	78 supply rd	Delhi
Keyboard	3	S1	cats	12 main st	Delhi
Monitor	15	S2	electrotech	45 tech ave	Mumbai
Keyboard	2	S2	electrotech	45 tech ave	Mumbai

b) List the name ,city, and address of the suppliers who are supplying keyboard.

SQL> select distinct s.sname,s.scity,s.saddress from suplier s join computer_items ci on s.supplier_no=ci.supplier_no where lower(ci.item_name)='keyboard';

SNAME	SCITY	SADDRESS
Cats	Delhi	12 main st
Electrotech	Mumbai	45 tech ave

c) List the supplier's name, items supplied by the suppliers 'Cats' and 'Electrotech'.

SQL> select s.sname,ci.item_name from suplier s join computer_items ci on s.supplier_no=ci.supplier_no where s.sname in('cats','electrotech');

SNAME	ITEM_NAME
Cats	Mouse
Cats	Keyboard
Electrotech	Monitor
Electrotech	Keyboard

d) Find the items having quantity less than 5 and insert the details of supplier and item of these, into another table NEWORDER

SQL> create table neworder(

2 supplier_no varchar(10),

3 item_no varchar(10),

4 item_name varchar(50),

5 Iquantity number(5),

6 Foreign key(supplier_no)references suplier(supplier_no));

Table created.

SQL> insert into neworder(supplier_no,item_no,item_name,Iquantity) select ci.supplier_no,ci.item_no,ci.item_name,ci.Iquantity from computer_items ci where ci.Iquantity<5;

2 rows created.

```
SQL> select * from neworder;

SUPPLIER_N      ITEM_NO      ITEM_NAME      IQUANTITY
    S1            I2         Keyboard           3
    S2            I4         Keyboard           2
```

Program 7:

Create the following tables by identifying primary and foreign keys, specify the not null property for mandatory keys.

PRODUCT_DETAIL				
P_NO	**PRODUCTNAME**	**QTYAVAILABLE**	**PRICE**	**PROFIT %**
P0001	Monitor	10	3000	20
P0002	Pen Drives	50	650	5
P0003	CD Drive	100	10	3
P0004	Key Board	25	600	10

PURCHASED_DETAIL		
CUSTNO	**P_NO**	**QTYSOLD**
C1	P0003	2
C2	P0002	4
C3	P0002	10
C4	P0001	3
C1	P0004	2
C2	P0003	2
C4	P0004	1

Write the following SQL queries:

a) Display total amount spent by C2.

b) Display the names of product for which either QtyAvailable is less than 30 or total QtySold is less than 5(USE UNION).

c) Display the name of products and quantity purchased by C4.

d) How much Profit does the shopkeeper gets on C1's purchase?

e) How many 'Pen Drives' have been sold?

```
SQL> create table product_detaill(
  2  p_no varchar(10)primary key,
  3  productname varchar(50)not null,
  4  qtyavailable number not null,
  5  price number not null,
  6  profit_percent number);


Table created.
```

```
SQL> desc product_detaill;
 Name                                      Null?              Type
 ----------------------------------------- -------- ----------------------------
 P_NO                                      NOT NULL           VARCHAR2(10)
 PRODUCTNAME                               NOT NULL           VARCHAR2(50)
 QTYAVAILABLE                              NOT NULL           NUMBER
 PRICE                                     NOT NULL           NUMBER
 PROFIT_PERCENT                                               NUMBER
```

SQL> create table purchased_detail(

```
  2  custno varchar(10) not null,
  3  p_no varchar(10) not null,
  4  qtysold number not null,
  5  foreign key(p_no)references product_detaill(p_no));

Table created.

SQL> desc purchased_detail;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 CUSTNO                                    NOT NULL VARCHAR2(10)
 P_NO                                      NOT NULL VARCHAR2(10)
 QTYSOLD                                   NOT NULL NUMBER

SQL> insert into product_detaill values('p0001','monitor',10,30000,20);
1 row created.
SQL> insert into product_detaill values('p0002','pendrives',50,650,5);
1 row created.
SQL> insert into product_detaill values('p0003','CD drive',100,10,3);
```

1 row created.

SQL> insert into product_detaill values('p0004','keyboard',25,600,10);

1 row created.

SQL> select * from product_detaill;

P_NO	PRODUCTNAME	QTYAVAILABLE	PRICE	PROFIT_PERCENT
p0001	monitor	10	30000	20
p0002	pendrives	50	650	5
p0003	CD drive	100	10	3
p0004	keyboard	25	600	10

SQL> insert into purchased_detail values('c1','p0003',2);

1 row created.

SQL> insert into purchased_detail values('c2','p0002',4);

1 row created.

SQL> insert into purchased_detail values('c3','p0002',10);

1 row created.

SQL> insert into purchased_detail values('c4','p0001',3);

1 row created.

SQL> insert into purchased_detail values('c1','p0004',2);

1 row created.

SQL> insert into purchased_detail values('c2','p0003',2);

1 row created.

SQL> insert into purchased_detail values('c4','p0004',1);

1 row created.

SQL> select * from purchased_detail;

CUSTNO	P_NO	QTYSOLD
c1	p0003	2
c2	p0002	4
c3	p0002	10
c4	p0001	3
c1	p0004	2
c2	p0003	2
c4	p0004	1

7 rows selected.

a) Display total amount spent by C2.

```
SQL> select custno,sum(qtysold*price) as total_amount_spent
    from purchased_detail pd
    join product_detaill p on pd.p_no=p.p_no
    where custno='c2'
   group by custno;
```

CUSTNO	TOTAL_AMOUNT_SPENT
c2	2620

b) Display the names of product for which either QtyAvailable is less than 30 or total QtySold is less than 5(USE UNION).

```
SQL> select productname from product_detaill where qtyavailable<30
    union
    select p.productname
    from product_detaill p
    join purchased_detail pd on p.p_no=pd.p_no
    group by p.productname
    having sum(qtysold)<5;
```

PRODUCTNAME

CD drive

keyboard

monitor

c) Display the name of products and quantity purchased by C4.

SQL> select p.productname,pd.qtysold from purchased_detail pd join product_detaill p on pd.p_no=p.p_no where pd.custno='c4';

PRODUCTNAME	QTYSOLD
monitor	3
keyboard	1

d) How much Profit does the shopkeeper gets on C1's purchase?

SQL> select sum((p.price*pd.qtysold)*(p.profit_percent/100)) as total_profit from purchased_detail pd join product_detaill p on pd.p_no=p.p_no where pd.custno='c1';

TOTAL_PROFIT

120.6

e) How many 'Pen Drives' have been sold?

SQL> select sum(pd.qtysold)as total_pen_drives_sold from purchased_detail pd join product_detaill p on pd.p_no=p.p_no where p.productname='pendrives';

TOTAL_PEN_DRIVES_SOLD

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